Demetrius Gulewicz

574-606-3840
dgulewic@purdue.edu

EDUCATION

Ph.D., Mechanical Engineering, Purdue University, West Lafayette, IN — Expected Spring 2028
Advisor: Dr. Neera Jain — GPA: 3.82/4.00

B.S., Mechanical Engineering, Purdue University, West Lafayette, IN — Spring 2023
GPA: 3.78/4.00

RESEARCH INTERESTS

Dynamic modeling, Thermal Systems, Adaptive and Optimal Control, Global Constrained System Optimization

HONORS & AWARDS

1. Herrick Assistantship, Ray W. Herrick Laboratories, 2025 – 2026
2. Best Poster Finalist, IEEE ITHERM Conference, 2024
3. HDG and CV Scheffer Scholarship, 2021 – 2023
4. Grant Arrasmith Mechanical Engineering Scholarship, 2020 – 2021
5. W Coleman Atkins Alumni Scholarship, 2020 – 2022
6. Gordon A Baugh Sr. Engineering Scholarship, 2019 – 2021

PUBLICATIONS

Peer Reviewed Journal Articles

1. **Gulewicz, D.** and Jain, N. "On Improving the Numerical Robustness of Charge-Sensitive Dynamic Refrigeration Models." Internation Journal of Refrigeration, August 19, 2026.

2. **Gulewicz, D.**, Inyang-Udoh, U., Bird, T. J., and Jain, N. "Nonlinear Model Predictive Control of a Hybrid Thermal Management System." IEEE Transactions on Control Systems Technologies, December 30, 2025.

3. **Gulewicz, D.**, Bird, T. J., Pangborn, H. C., and Jain, N. "Set-Based Robust Control Co-Design: Application to a Hybrid Thermal Management System." ASME. *Letters Dyn. Sys. Control.* July 2025; 5(3).

Peer Reviewed Conference Papers

1. Dewanatha, P. P., **Gulewicz, D.** and Jain, N. (October 14, 2024) "Closed-loop Analysis of Thermal Energy Storage Device Arrangement in a Thermal Management System." *2024 23rd IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (ITherm).* Aurora, CO, USA. 2024. pp. 1-10.

RESEARCH EXPERIENCE

Graduate Research Assistant, Ray W. Herrick Labs, Purdue University — May 2023 - Present

- Designed and implemented a real-time non-linear model predictive control scheme on a single-phase cooling system
- Updated physical testbed of a single-phase cooling system augmented with thermal energy storage, involving re-design, procurement, manufacturing, and assembly
- Developed novel model formulations of vapor compression dynamics to reduce computation time and improve robustness; implemented the formulations into a Python toolbox

TEACHING EXPERIENCE

ME 375: Systems, Measurements, and Controls II, Graduate Teaching Assistant — July 2025 – December 2025

- Developed new laboratory assignments as part of a course re-design effort

- Tutored students in lab and in class on key controls concepts

LEADERSHIP EXPERIENCE

Research Mentor June 2024 – Present

- Mentored a student in the Purdue SURF program to develop an optimal control strategy for the control of mass flow rate in a single phase, branched fluid circuit. Awarded best poster at the SURF symposium
- Mentored a student under the Purdue SURF program to develop a control-oriented dynamic building model
- Both students continued after the end of the SURF program to work on further research objectives

Purdue Electric Racing Controls Sub-Team Mentor June 2025 - Present

- Mentored new sub-team lead on resource allocation, prioritization, technical skills, and general leadership
- Mentored a student to construct a suspension kinematics solver for use as a design tool for the vehicle dynamics sub-team. The kinematics solver uses a trilateration algorithm to resolve all suspension hard points
- Oversaw the development of a driver-in-loop simulator using a fully transient 6DOF vehicle model that can run in real-time on an ordinary laptop though a Simulink-Hardware interface

Purdue Electric Racing Controls Sub-Team Leader May 2022 - May 2025

- Founder of the controls sub-team at Purdue Electric Racing
- Coordinated torque vectoring implementation across sub-teams to meet system requirements
- Successfully implemented torque vectoring to increase steady state lateral acceleration
- Formulated and successfully implemented a custom 6DOF transient vehicle model to facilitate model-based design optimization across sub-teams

WORK EXPERIENCE

Student Intern at Vehicle Propulsion Systems, Ford Research & Innovation Center May 2026 – August 2026

- Developed novel optimization formulation to efficiently solve select mixed-integer nonlinear programs (MINLPs).
- Proposed integrated optimization approach for use in architecture selection and control design.
- Analyzed vehicle thermal architectures for potential improvements to energy efficiency and controllability.

Student Associate at Honda Powersports, Honda R&D Americas May 2022 – August 2022

- Clarified control specifications for a future vehicle platform in a presentation to management to guide project workflow
- Tabulated standard testing procedure & pass criteria for rechargeable energy storage systems to advance testing readiness
- Investigated market solutions that can be benchmarked against proposed vehicle architecture, leading to 1 vehicle purchase
- Proposed test bench hardware and software to enable agile research and development